

SI 140A.02 Probability & Statistics for EECS, Fall 2026

Homework 3

Name:

Student ID:

Due date: Due on Oct. 11, 2026, 23:59 UTC+8

Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.
- You are required to submit your homework in PDF format generated by LaTeX, otherwise you will get no points of this homework.

Problem 1

There are two states, Red and Blue, initially with 100 residents each. Every resident supports exactly one of two parties, Democrat or Republican. Initially, 60 residents of Blue and 30 residents of Red are Democrats. Ten residents move from Blue to Red, and nobody changes parties. Let k be the number of Democrats among the ten people who move.

Whenever a resident is selected, the selection is uniform over all 200 people. Let D be the event that the selected resident is a Democrat, and let B be the event that the resident lives in Blue at the time of selection.

- Find $P(D | B)$, $P(D | B^c)$, and $P(D)$ before the move. Does the move change $P(D)$? Explain.
- Determine all integer values of k for which the proportion of Democrats increases strictly in *both* states. Explain, using the law of total probability with the appropriate weights, how both proportions can increase while the overall proportion stays unchanged.
- For this part, take $k = 5$ and consider the population after the move. A survey interviews the selected resident only if their name appears on a list containing all Blue residents and all Democrats in Red. Let A be the event that the selected resident is on this list. Find $P(D | A)$ using the law of total probability conditional on A , with the partition $\{B, B^c\}$, and state the two group weights. Explain why this probability differs from the proportion of Democrats in the whole population.

Solution

- Before the move, $P(B) = P(B^c) = 1/2$. Hence

$$P(D | B) = \frac{60/200}{1/2} = \frac{3}{5}, \quad P(D | B^c) = \frac{30/200}{1/2} = \frac{3}{10}, \quad P(D) = \frac{90}{200} = \frac{9}{20}.$$

The move does not change $P(D)$. Nobody changes parties, so the total number of Democrats remains 90. The population remains 200, and selection is uniform over these 200 residents. Thus the overall probability remains $9/20$.

- After the move, Blue has 90 residents, including $60 - k$ Democrats, and Red has 110 residents, including $30 + k$ Democrats. Therefore,

$$P(D | B) = \frac{60 - k}{90}, \quad P(D | B^c) = \frac{30 + k}{110}.$$

Both proportions increase strictly if and only if

$$\frac{60 - k}{90} > \frac{3}{5} \iff k < 6, \quad \frac{30 + k}{110} > \frac{3}{10} \iff k > 3.$$

Thus the possible integer values are $k = 4$ or 5 . By the law of total probability, using the post-move population weights,

$$P(D) = P(D | B)P(B) + P(D | B^c)P(B^c) = \frac{60 - k}{90} \frac{90}{200} + \frac{30 + k}{110} \frac{110}{200} = \frac{9}{20}.$$

Although both state proportions increase, the weights change from $(1/2, 1/2)$ to $(9/20, 11/20)$. The higher-proportion state, Blue, receives less weight, while Red receives more weight, leaving the overall proportion unchanged.

- (c) With $k = 5$, Blue contains 55 Democrats among 90 residents, and Red contains 35 Democrats among 110 residents. The list contains all 90 Blue residents and the 35 Democrats in Red, for a total of 125 residents. The two conditional group weights are

$$P(B \mid A) = \frac{90}{125} = \frac{18}{25}, \quad P(B^c \mid A) = \frac{35}{125} = \frac{7}{25}.$$

Within these groups, $P(D \mid A, B) = 55/90 = 11/18$ and $P(D \mid A, B^c) = 1$. By the law of total probability conditional on A ,

$$\begin{aligned} P(D \mid A) &= P(D \mid A, B)P(B \mid A) + P(D \mid A, B^c)P(B^c \mid A) \\ &= \frac{11}{18} \frac{18}{25} + 1 \cdot \frac{7}{25} = \boxed{\frac{18}{25}}. \end{aligned}$$

This exceeds the overall proportion $9/20$ because the list includes every Democrat but excludes all 75 Republicans in Red. Conditioning on being listed therefore increases the proportion of Democrats.

Problem 2

A sequence of $n \geq 1$ mutually independent trials is performed. Each trial results in either success or failure, and the probability of success on trial i is p_i , where $0 \leq p_i \leq 1$. Let A_n be the event that the number of successes among the first n trials is even.

- (a) Find $P(A_2)$ by listing the possible outcomes, and show that

$$P(A_2) = \frac{1 + (1 - 2p_1)(1 - 2p_2)}{2}.$$

- (b) By considering whether the last trial succeeds or fails, obtain a recursion for $P(A_n)$ for $n \geq 2$. Use the recursion to prove that

$$P(A_n) = \frac{1 + \prod_{i=1}^n (1 - 2p_i)}{2}.$$

You may use mathematical induction or repeated substitution in the recursion. State the initial value explicitly.

- (c) Under the independence assumption, is $P(A_n) = 1/2$ if and only if at least one $p_i = 1/2$? Justify your answer. Then remove the independence assumption and specify probabilities for the four possible outcomes of two trials so that $p_1 = p_2 = 1/2$ but $P(A_2) \neq 1/2$.

Solution

- (a) Let S_i be the event that trial i succeeds, and write S and F for success and failure in an ordered outcome. The outcomes in A_2 are SS and FF , with probabilities p_1p_2 and $(1 - p_1)(1 - p_2)$, respectively. Thus, for all $0 \leq p_1, p_2 \leq 1$,

$$P(A_2) = P(SS) + P(FF) = p_1p_2 + (1 - p_1)(1 - p_2).$$

When $0 < p_1 < 1$, this can also be written by conditioning on the first trial:

$$\begin{aligned} P(A_2) &= P(A_2 | S_1)P(S_1) + P(A_2 | S_1^c)P(S_1^c) \\ &= p_2p_1 + (1 - p_2)(1 - p_1) \\ &= 1 - p_1 - p_2 + 2p_1p_2 = \boxed{\frac{1 + (1 - 2p_1)(1 - 2p_2)}{2}}. \end{aligned}$$

At $p_1 = 0$ or 1 , the same result follows directly from the outcome sum above.

- (b) If trial n fails, the total number of successes is even exactly when A_{n-1} occurs. If trial n succeeds, it is even exactly when A_{n-1}^c occurs. Independence and the law of total probability give

$$\boxed{P(A_n) = (1 - p_n)P(A_{n-1}) + p_n(1 - P(A_{n-1}))}, \quad n \geq 2.$$

The initial value is $P(A_1) = 1 - p_1$, since zero successes is even. This agrees with the claimed formula:

$$P(A_1) = 1 - p_1 = \frac{1 + (1 - 2p_1)}{2}.$$

For the induction step, assume that

$$P(A_k) = \frac{1 + \prod_{i=1}^k (1 - 2p_i)}{2}.$$

Then the recursion yields

$$\begin{aligned} P(A_{k+1}) &= (1 - p_{k+1})P(A_k) + p_{k+1}(1 - P(A_k)) \\ &= p_{k+1} + (1 - 2p_{k+1})P(A_k) \\ &= p_{k+1} + (1 - 2p_{k+1})\frac{1 + \prod_{i=1}^k (1 - 2p_i)}{2} \\ &= \frac{1 + \prod_{i=1}^{k+1} (1 - 2p_i)}{2}. \end{aligned}$$

Therefore, by mathematical induction, the formula holds for every $n \geq 1$.

(c) Under independence, the formula in part (b) gives

$$P(A_n) = \frac{1}{2} \iff \prod_{i=1}^n (1 - 2p_i) = 0 \iff p_i = \frac{1}{2} \text{ for at least one } i.$$

The last equivalence holds because a finite product is zero if and only if at least one factor is zero.

Without independence, consider the joint distribution

Outcome	<i>SS</i>	<i>SF</i>	<i>FS</i>	<i>FF</i>
Probability	1/2	0	0	1/2

Then $p_1 = p_2 = 1/2$, but

$$P(A_2) = P(SS) + P(FF) = 1 \neq \frac{1}{2}.$$

The trials are dependent: $P(SS) = 1/2 \neq p_1 p_2 = 1/4$.

Problem 3

To battle against spam, Bob installs two anti-spam programs. An email is either legitimate (event L) or spam (event L^c). Let M_j be the event that program j marks the email as legitimate, for $j = 1, 2$. Assume that

$$P(L) = 0.10, \quad P(M_j | L) = P(M_j^c | L^c) = 0.90 \quad (j = 1, 2).$$

The two programs process the same email. Their outputs are conditionally independent given L , and also conditionally independent given L^c .

- Find $P(L | M_1)$ and simplify your answer.
- Find $P(L | M_1, M_2)$ and $P(L | M_1, M_2^c)$. Compare each with the prior probability $P(L)$.
- Are M_1 and M_2 independent without conditioning on the email's true status? Prove your answer by computing the required probabilities. Explain why your answer is consistent with the stated conditional independence.

Solution

- First, taking complements gives

$$P(M_j | L^c) = 1 - P(M_j^c | L^c) = 0.10, \quad j = 1, 2.$$

By the law of total probability,

$$P(M_1) = P(M_1 | L)P(L) + P(M_1 | L^c)P(L^c) = 0.90 \cdot 0.10 + 0.10 \cdot 0.90 = 0.18.$$

Thus Bayes' rule gives

$$P(L | M_1) = \frac{P(M_1 | L)P(L)}{P(M_1)} = \frac{0.90 \cdot 0.10}{0.18} = \boxed{\frac{1}{2}}.$$

- Using conditional independence and the law of total probability,

$$P(M_1 \cap M_2) = 0.90^2 \cdot 0.10 + 0.10^2 \cdot 0.90 = 0.09.$$

Therefore, by Bayes' rule,

$$P(L | M_1, M_2) = \frac{P(M_1 \cap M_2 | L)P(L)}{P(M_1 \cap M_2)} = \frac{0.90^2 \cdot 0.10}{0.09} = \boxed{\frac{9}{10}}.$$

Similarly, conditional independence also applies when the second event is replaced by its complement, so

$$P(M_1 \cap M_2^c) = 0.90 \cdot 0.10 \cdot 0.10 + 0.10 \cdot 0.90 \cdot 0.90 = 0.09.$$

Hence

$$P(L | M_1, M_2^c) = \frac{P(M_1 \cap M_2^c | L)P(L)}{P(M_1 \cap M_2^c)} = \frac{0.90 \cdot 0.10 \cdot 0.10}{0.09} = \boxed{\frac{1}{10}}.$$

When both programs mark the email as legitimate, the posterior 0.90 exceeds the prior $P(L) = 0.10$. When the programs disagree, the posterior equals the prior.

- By the law of total probability, for $j = 1, 2$,

$$P(M_j) = 0.90 \cdot 0.10 + 0.10 \cdot 0.90 = 0.18.$$

However, part (b) gives

$$P(M_1 \cap M_2) = 0.09 \neq 0.18^2 = 0.0324 = P(M_1)P(M_2).$$

Thus M_1 and M_2 are not independent without conditioning. This is consistent with conditional independence: within each fixed email status, the outputs are independent, but both programs are more likely to mark a legitimate email as legitimate than a spam email. Averaging over the shared, unknown status introduces dependence between their outputs.

Problem 4

Consider a simplified model of an AI evaluation. In one run, an AI system answers two questions. Write C for correct and I for incorrect; the ordered outcomes CC, CI, IC, II also denote the corresponding events. In parts (a) and (b), each answer is correct with probability $1/2$, independently of the other, so the four outcomes are equally likely.

An evaluator checks both answers against an answer key and may publish one of the following reports. Let M_C and M_I denote the events that the corresponding reports are published:

$$\begin{aligned} M_C : & \text{ "At least one answer is correct";} \\ M_I : & \text{ "At least one answer is incorrect".} \end{aligned}$$

At most one report is published per run, and every published report must be true.

- (a) Initially, every run is reported. For CC , the evaluator publishes M_C ; for II , the evaluator publishes M_I . For either CI or IC , a fair coin is used to choose between the two reports, each with probability $1/2$. Using Bayes' rule, find

$$P(CC | M_C) \quad \text{and} \quad P(II | M_I).$$

- (b) Replace the rule in part (a) by any randomized reporting rule based on the outcome. The evaluator may choose not to publish a report for any of the four outcomes. The two reports must each have positive probability and must satisfy

$$P(CC | M_C) \geq \frac{2}{3}, \quad P(II | M_I) \geq \frac{3}{4}.$$

What is the smallest possible probability that no report is published? Specify a reporting rule for each of CC, CI, IC, II that attains your answer, and prove that no admissible rule can do better.

- (c) Remove the independence assumption, but keep each answer's probability of being correct equal to $1/2$. The evaluator may choose a new randomized reporting rule, but must publish exactly one truthful report per run, with both report types having positive probability. Let $t = P(CC)$. Determine all values of t for which such a rule satisfies both posterior inequalities in part (b). Prove necessity and sufficiency, giving a valid rule for every claimed value of t . The rule may depend on t .

Solution

- (a) By the law of total probability,

$$P(M_C) = 1 \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{2}.$$

Similarly, $P(M_I) = 1/2$. Bayes' rule therefore gives

$$P(CC | M_C) = \frac{P(M_C | CC)P(CC)}{P(M_C)} = \frac{1 \cdot (1/4)}{1/2} = \boxed{\frac{1}{2}},$$

$$P(II | M_I) = \frac{P(M_I | II)P(II)}{P(M_I)} = \frac{1 \cdot (1/4)}{1/2} = \boxed{\frac{1}{2}}.$$

(b) Let N be the event that no report is published. The posterior constraints imply

$$P(M_C) \leq \frac{3}{2}P(CC \cap M_C) \leq \frac{3}{2}P(CC) = \frac{3}{8},$$

$$P(M_I) \leq \frac{4}{3}P(II \cap M_I) \leq \frac{4}{3}P(II) = \frac{1}{3}.$$

Since the reports are mutually exclusive,

$$P(N) = 1 - P(M_C) - P(M_I) \geq 1 - \frac{3}{8} - \frac{1}{3} = \frac{7}{24}.$$

To attain this bound, use the following rule. Each row gives the conditional probabilities of the three reporting choices given the outcome.

Outcome	M_C	M_I	No report
CC	1	0	0
CI	1/4	1/6	7/12
IC	1/4	1/6	7/12
II	0	1	0

Every published report is true. For this rule,

$$P(M_C) = \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} = \frac{3}{8}, \quad P(M_I) = \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{6} = \frac{1}{3}.$$

Thus both report types have positive probability, and

$$P(CC | M_C) = \frac{1/4}{3/8} = \frac{2}{3}, \quad P(II | M_I) = \frac{1/4}{1/3} = \frac{3}{4}.$$

Finally, $P(N) = (1/2)(7/12) = 7/24$. Hence the smallest possible probability of publishing no report is $7/24$.

(c) The marginal probabilities of correctness imply

$$P(CI) = \frac{1}{2} - t, \quad P(IC) = \frac{1}{2} - t, \quad P(II) = t.$$

Nonnegativity requires $0 \leq t \leq 1/2$.

Necessity. Exactly one truthful report must be published per run. Consequently, CC must produce M_C , and II must produce M_I , so

$$P(CC \cap M_C) = P(II \cap M_I) = t.$$

The posterior constraints give

$$P(M_C) \leq \frac{3}{2}t, \quad P(M_I) \leq \frac{4}{3}t.$$

Since $P(M_C) + P(M_I) = 1$, it follows that

$$1 \leq \left(\frac{3}{2} + \frac{4}{3}\right)t = \frac{17}{6}t, \quad \text{so} \quad t \geq \frac{6}{17}.$$

Sufficiency. For any $6/17 \leq t \leq 1/2$, use the rule

Outcome	M_C	M_I
CC	1	0
CI	3/5	2/5
IC	3/5	2/5
II	0	1

Each row gives conditional reporting probabilities. The rule always publishes exactly one truthful report, and

$$P(M_C) = t + \frac{3}{5}(1 - 2t) = \frac{3 - t}{5}, \quad P(M_I) = t + \frac{2}{5}(1 - 2t) = \frac{2 + t}{5}.$$

Both probabilities are positive throughout the claimed interval. Moreover,

$$P(CC \mid M_C) = \frac{5t}{3 - t} \geq \frac{2}{3} \iff 17t \geq 6,$$

$$P(II \mid M_I) = \frac{5t}{2 + t} \geq \frac{3}{4} \iff 17t \geq 6.$$

Thus both requirements hold, including at the endpoints. The set of all feasible values is therefore $t \in [6/17, 1/2]$.